Supplementary material

A Model parameterization and algebraic properties

The frozen ESM-2 model supplies 1,280-dimensional residue representations. Independent linear projections map wild-type, local-change, and cross-site-change features to the selected projection dimension. Amino-acid states use one-hot encodings. The single and pair heads have a 256-dimensional hidden layer, a 128-dimensional second layer, GELU activations, and a scalar output. The pair head also receives the calibrated single effects and their observation indicators. Its sequence relation uses the logarithm of one plus the site gap and the gap divided by sequence length. Projection dimensions, learning rates, auxiliary weights, and ω are selected from source-validation measurements.

The cross-site context is the change at one site after the other site is substituted,

$$\chi_{i\leftarrow j}(b) = h_i(x_p(j,b)) - h_i(s_p). \tag{11}$$

Each single-mutant sequence is encoded once. Its residue states are reused across candidate pairs. The pair network therefore receives mutation-dependent context without encoding every double-mutant sequence. Sequence separation supplies the relation between endpoints. The structural variant adds reference Cα distances and residue confidence.

The normalized phenotype map is

$$g(z) = \frac{z + \sum_{k=1}^{20} b_k[\tanh(c_k z + d_k) - \tanh(d_k)]}{1 + \sum_{k=1}^{20} b_k c_k \operatorname{sech}^2(d_k)}, \qquad b_k, c_k > 0. \tag{12}$$

Positive parameters use softplus. Substitution of $z = 0$ gives $g(0) = 0$. Differentiation gives $g'(0) = 1$. Its derivative is positive for every z, and its linear term preserves nonzero slopes in the tails. The positive factor τ sets the phenotype scale.

For $u = \widetilde{a}_i(a)$ and $v = \widetilde{a}_j(b)$, the predicted phenotype contrast decomposes as

$$\widehat{\epsilon}_{\text{global}} = \tau[g(u+v) - g(u) - g(v)], \tag{13}$$

$$\widehat{\epsilon}_{\text{pair}} = \tau[g(u+v+r_{ij}) - g(u+v)]. \tag{14}$$

Their sum is $\widehat{\epsilon}_p$. With the identity map and unit scale, the phenotype contrast equals r_{ij}. With a nonlinear map, the complete contrast includes both contributions.

Symmetry of F implies symmetry of the cycle subtraction. Setting $b = w_j$ makes its first two terms equal and its last two terms equal, giving $r_{ij}(a, w_j) = 0$. The other reference constraint follows identically. For an endpoint term $v_i(a)$, the subtraction contains $v_i(a) - v_i(a) - v_i(w_i) + v_i(w_i) = 0$. The second endpoint term and the constant cancel in the same way. These are algebraic properties of the parameterization. Real-data model checks also verify endpoint exchange and reference-state outputs.

Matched controls for the pair function

The input-removal control supplies model-predicted single effects $a_i(a), a_j(b)$ to every evaluation of F and sets its observation indicators to zero. Equation (3) continues to define the external single-effect sum. Measured values and their availability indicators therefore reach that sum through the original permission rule. The pair network receives neither channel. Its input dimensions and parameter count remain unchanged.

The direct-head control predicts $r_{ij}(a, b) = F(i, a; j, b)$ with the same symmetric network. Its projections, endpoint features, measured inputs, hidden dimensions, and parameter count match SCEPI. This intervention isolates cycle subtraction while retaining endpoint symmetry and the external single-effect calibration.

Both controls reuse the full model's source-selected hyperparameters in each outer partition. They use the same pretrained single checkpoint for each training repetition, random initialization, source rows, sampler, optimizer, objectives, and checkpoint-selection rule. The maximum epoch budget and early-stopping patience are identical. Realized stopping epochs and fitted weights may differ. The label-mixing weight follows the same source-validation procedure. Full-target evaluation retains the same proteins, candidate pools, measurement margins, budgets, and aggregation. The stricter homology grouping repeats this comparison independently. These controls measure the predictive contributions of cycle subtraction and measured pair inputs.

B Repetitions, aggregation, and statistical analysis

The 30% grouping has five outer partitions. Each source-trained model uses one partition for testing, the next partition for source validation, and the remaining three for source training. The main comparisons contain all 25 outer-fold and training-seed evaluations per model. Training seeds are 11, 23, 47, 89, and 131. Every eligible target protein is evaluated once per training seed in its assigned outer test partition. All eligible target double mutants enter the full-target evaluation. There are 795 paired protein-repetition records for phenotype prediction, 775 for residual prediction, and 600 for standard compensation selection. The 0.5 kcal/mol margin cohort contains 575 records. Each protein is averaged over its five repetitions before the equal-protein mean is computed.

At each repetition, all compared methods use the same permitted target labels and candidate definition. Means use the paired cohort. Phenotype and residual comparisons cover 159 and 155 proteins, respectively, from 51 homology groups. Standard compensation covers 120 proteins and 44 groups. The 0.5 kcal/mol margin comparison covers 115 proteins and 43 groups. Source-validation configurations are retained across training repetitions. Candidate yield is excluded from source hyperparameter selection. Five outer partitions describe source-data separation; five training seeds describe repeated model fitting.

Uncertainty uses 10,000 homology-group bootstrap draws with statistical seed 20261013. Each draw samples groups with replacement and retains all member proteins. Its effect is the equal-protein mean over the sampled groups. Repeated model fits are averaged within protein before resampling. Comparisons with the same group membership use the same resampling draws. A bootstrap positive fraction is the fraction of resampled mean differences above zero. It describes uncertainty in the observed paired mean under this resampling scheme. Tables 4 and 5 report the top-10 endpoint. Table 6 reports the matched pair-function controls.

Table 4. Paired native top-10 hit-rate advantage of SCEPI. Gains and 95% bootstrap intervals are in percentage points. Positive fractions refer to resampled mean gains. Every comparison includes 120 proteins, 44 groups, and five model repetitions.

Comparator	Mean gain	95% interval	Positive fraction
Additive	4.83	[1.97, 7.95]	0.9991
Global nonlinear	6.77	[3.10, 10.11]	1.0000
ThermoMPNN-D	4.15	[−0.32, 7.08]	0.9628
Kermut	5.82	[2.59, 8.79]	0.9996
StabilityArc + Kermut	2.47	[−1.17, 5.55]	0.9165
Mean MLP	1.15	[−2.90, 3.76]	0.7387
Mean ridge	−0.20	[−4.10, 1.92]	0.4312

Table 5. Paired top-10 hit-rate advantage with the same measured single-effect sum at margin $\geq$ 0.5 kcal/mol. Gains and 95% bootstrap intervals are in percentage points. Every comparison includes 115 proteins, 43 groups, and five model repetitions.

Comparator	Mean gain	95% interval
Additive	6.63	[2.79, 9.96]
Global nonlinear	6.33	[2.38, 9.39]
ThermoMPNN-D	5.88	[2.39, 8.55]
Kermut	5.86	[2.19, 9.15]
StabilityArc + Kermut	3.03	[−0.64, 6.41]
Mean MLP	3.84	[0.27, 5.76]
Mean ridge	3.67	[0.77, 6.13]

Table 6. Paired advantage of SCEPI over matched pair-function controls under the 30% and 25% identity groupings. Residual RMSE reductions are in kcal/mol; native top-10 hit-rate gains are in percentage points. Brackets give 95% homology-group bootstrap intervals. Residual and selection comparisons contain 155 and 120 proteins, respectively. Each protein has five matched repetitions.

Grouping	Pair-function control	Residual reduction	Hit-rate gain
30%	Predicted singles	0.172 [0.152, 0.211]	3.03 [0.72, 4.32]
30%	Direct residual	0.041 [−0.014, 0.073]	0.52 [−3.04, 2.46]
25%	Predicted singles	0.210 [0.155, 0.263]	3.00 [−0.06, 4.36]
25%	Direct residual	0.026 [−0.019, 0.042]	−0.23 [−1.58, 0.81]

Measured pair inputs improve both mean residual error and native hit rate in all five repetitions under each grouping. Their residual comparison has 143 protein-level wins under the primary grouping and 142 under the stricter grouping, out of 155 proteins. The reference contrast lowers mean residual error in all five repetitions at each threshold, with 106 and 97 protein-level wins. Its native hit-rate gain is positive in four repetitions under the 30% grouping and two under the 25% grouping.

C Source model selection and implementation

Complete source cycles provide the residual target in Equation (1). For source proteins P, the residual objective gives equal weight to proteins and site pairs,

$$\mathcal{L}_\epsilon = \frac{1}{|P|}\sum_{p\in P}\frac{1}{|J_p|}\sum_{(i,j)\in J_p}\frac{1}{|Q_{pij}|}\sum_{q\in Q_{pij}}\mathrm{Huber}\left(\frac{\widehat{\epsilon}_q-\epsilon_q}{s}\right), \tag{15}$$

where J_p contains measured site pairs and Q_{pij} contains their cycles. The stability scale is $s = 1$ kcal/mol. Sampling across proteins, then pairs, then cycles implements this weighting.

The fitted objective is $\mathcal{L}_\epsilon + \lambda_y \mathcal{L}_y$. The observation term $\mathcal{L}_y$ is the negative log-likelihood for the predicted phenotype. Point observations use a Gaussian density; interval observations use the corresponding Gaussian interval probability. Each original observation appears once in this term. This criterion jointly fits the single head, pair function, and phenotype map. Residual-only and phenotype-only models are trained separately.

The single-effect model is fitted to source single-mutant measurements before joint training. The joint configuration grid contains projection dimensions of 64 and 128, learning rates of 10^{-4}, 3×10^{-4}, and 10^{-3}, and auxiliary phenotype weights of 0.1 and 1. The resulting twelve configurations are fitted separately in each outer partition. AdamW uses weight decay 10^{-4} and batches of 1,024 records. Training runs for at most 100 epochs, with early stopping after ten epochs without validation improvement. The residual loss uses Huber threshold 1 on the kcal/mol scale.

Training samples three single-label availability conditions. With probability 0.5, it hides target single labels. With probability 0.25, it permits 128 single labels. With probability 0.25, it permits all single labels. Source validation selects $\omega \in \{0, 0.5, 1\}$ in the 128-label condition. It selects the model configuration by a weighted residual RMSE: 0.5 for zero labels, 0.25 for 128 labels, and 0.25 for all single labels. The selected configuration is then refitted for each training repetition. Phenotype-only, residual-only, and representation variants have separate fitted parameters. The additive model uses phenotype validation because its four-term residual is zero by construction.

Frozen sequence representations are cached by reference and single-mutant sequence. The pair predictor reuses those caches across double mutants. The structural variant adds Cα distance and residue confidence from the reference structure. Geometry permutations preserve the recorded distance values while changing their pair assignments. Inference interventions use the fitted structural model. Retrained interventions also use the changed assignments during fitting. Each comparison retains the same outer partition and information condition.

ThermoMPNN-D is retrained on source proteins with its double-mutant head and the published single-mutant data augmentation [3]. Its additive counterpart is trained separately. Kermut fits the permitted

target labels with the full composite kernel. Mean ridge and mean MLP fit those target labels using mean-pooled residue representations. ESM-2 additive and context scores provide fixed pretrained representation controls. The main external comparison evaluates each method's native predictions. Equation (10) replaces each method's single-effect sum with the same measured sum.

D StabilityArc architecture and target calibration

The StabilityArc comparison uses frozen ESMC-600M wild-type residue embeddings and attention-derived contacts [14]. Contact scores use symmetrization, average product correction (APC), and attention entropy weighting. Its decoder has four layers, hidden dimension 256, and eight attention heads with rotary position encoding. The full amino-acid embedding, sequence-distance features, and symmetric contact-aware pair residual are retained. Each of the ten source partitions is trained with the five fixed training seeds, giving 50 source models. Every source model excludes its target homology groups from both fitting and validation.

Each source model is trained for 40 epochs with AdamW, learning rate 10^{-4}, and weight decay 10^{-3}. Each assay contributes at most 40,000 measured variants per epoch. The objective combines Huber loss with 1,024 measured ranking pairs per assay. The Huber threshold is 1, and the two losses have equal weights. Double-mutant regression targets use the source-assay single-mutant median and median absolute deviation. Ranking uses measured stability order. Checkpoints are selected by source-validation double-mutant Spearman correlation. The public release supplies the architecture and a pooled checkpoint; the matched source-partition models are retrained here. Equal loss weights are an explicit implementation choice because the release does not specify them.

Target calibration uses the full Kermut covariance and Gaussian likelihood [6]. The retrained StabilityArc score replaces the prior mean input. All remaining kernel inputs and permitted target labels match the Kermut comparison. Target fitting uses 150 AdamW steps with learning rate 0.1 and weight decay 0.01. Only the permitted reference and single-mutant point labels are supplied. The posterior scores every eligible target double mutant and its constituent singles. The model's four-prediction contrast defines its residual.

E Component comparison details

Table 7. Top-10 compensation hit rates with the same measured single-effect sum. Each method contributes its own predicted four-term residual. The standard and margin cohorts contain 120 and 115 proteins, respectively.

Residual source	Standard pool (%)	Margin ≥ 0.5 (%)
SCEPI	**11.27**	**11.15**
ThermoMPNN-D	5.30	5.27
Kermut	5.47	5.29
StabilityArc + Kermut	8.17	8.12
Global nonlinear	4.70	4.82
Additive	4.42	4.52
Mean MLP	7.17	7.30
Mean ridge	7.33	7.48

Table 8. Full-target S-all results for predictors and model variants. RMSE and recovery are in kcal/mol. Hit rates are native top-10 compensation percentages. The phenotype, residual, and selection cohorts contain 159, 155, and 120 proteins. Each predictor is evaluated across five outer partitions and five training repetitions.

Predictor or variant	Phenotype	Residual	Hit rate	Recovery
SCEPI	0.657	0.595	11.22	−0.257
Additive	0.893	1.091	6.38	−0.387
Global nonlinear	0.911	0.635	4.45	−0.437
ThermoMPNN-D	0.777	0.805	7.07	−0.440
ThermoMPNN additive	0.967	1.091	5.30	−0.511
Kermut	0.798	0.847	5.40	−0.458
StabilityArc + Kermut	0.815	0.853	8.75	−0.344
Mean MLP	1.405	0.966	10.07	−0.413
Mean ridge	1.328	1.002	11.42	−0.354
ESM-2 additive	1.173	1.091	2.50	−0.794
ESM-2 context	1.107	0.864	8.50	−0.499
Pair removed	1.151	1.072	4.42	−0.432
Direct residual head	0.700	0.635	10.70	−0.302
Predicted singles	0.810	0.767	8.18	−0.396
Local context	0.637	0.575	11.47	−0.258
Wild-type context	0.643	0.573	10.53	−0.273
Identity features	0.617	0.557	9.12	−0.290
Phenotype loss	0.619	0.562	11.95	−0.229
Residual loss	1.071	0.686	8.72	−0.334
Frozen singles	0.689	0.619	9.75	−0.282
Identity map	0.651	0.593	11.13	−0.257
Structure	0.644	0.583	10.82	−0.263
Permuted distances^{I}	0.648	0.586	10.95	−0.261
Permuted distances^{R}	0.650	0.590	10.53	−0.270
Rewired distances^{I}	0.646	0.585	10.78	−0.264
Rewired distances^{R}	0.633	0.572	10.47	−0.257

Pair removal keeps the trained single-effect predictions and phenotype map fixed. The predicted-singles control replaces measured inputs inside F and retains measured single-effect calibration outside F. Local context omits cross-site changes. Wild-type context uses reference-sequence representations. Identity features use amino-acid identities and single effects, with hidden width adjusted to match the main model's parameter count. The remaining variants change the training loss, freeze the single-effect head, replace the phenotype map with the identity, or add structural distances. Superscripts I and R denote distance interventions during inference and retraining, respectively. Retrained additive and global nonlinear models optimize their own parameters.

Removing cross-site context yields 0.575 kcal/mol residual RMSE and an 11.47% native hit rate under the primary grouping. The amino-acid-identity version reaches 0.557 kcal/mol and 9.12%. Phenotype-only training reaches 0.562 kcal/mol and 11.95%. Under the stricter grouping, phenotype-only training lowers residual RMSE from 0.604 to 0.584 kcal/mol and raises hit rate from 11.15% to 11.98%. The identity phenotype map gives 0.593 kcal/mol and 11.13% under the primary grouping. Cross-site context, the flexible map, and the residual objective remain model and fitting choices with these individual results.

F Sensitivity to the homology threshold

Grouping at 25% identity with the same coverage requirement produces 45 groups in the phenotype and residual cohorts. The compensation cohort contains 39 groups, and its 0.5 kcal/mol margin subset contains 38. Protein membership remains unchanged. All predictors and components are evaluated across five outer

partitions and five training repetitions under this grouping (Table 9). SCEPI's residual RMSE is 0.604 kcal/mol, compared with 0.771 for ThermoMPNN-D, 0.847 for Kermut, 0.874 for StabilityArc + Kermut, and 0.650 for global nonlinearity. Its native compensation hit rate is 11.15%, compared with 6.97%, 5.40%, and 8.23% for the three external methods. Its mean selected recovery exceeds StabilityArc + Kermut by 0.092 kcal/mol.

With the same measured single-effect sum, top-10 hit rates are 11.17% for SCEPI, 4.42% for additivity, 4.63% for global nonlinearity, 5.68% for ThermoMPNN-D, 5.47% for Kermut, 8.57% for mean MLP, and 5.83% for mean ridge. At margin 0.5 kcal/mol, the corresponding rates are 11.23%, 4.52%, 4.75%, 5.65%, 5.29%, 8.77%, and 5.83%. The stricter grouping preserves the average compensation advantage with this fixed sum.

StabilityArc + Kermut reaches an 8.02% hit rate with the fixed single-effect sum and 7.95% at margin 0.5 kcal/mol. The paired comparison retains the same proteins and label permissions. SCEPI's selection advantage over this source-trained comparator persists under both homology thresholds.

The matched direct residual head reaches 0.630 kcal/mol residual RMSE and an 11.38% native hit rate. SCEPI lowers residual error to 0.604 kcal/mol and has an 11.15% native hit rate. Removing measured pair inputs while retaining the external calibration gives 0.814 kcal/mol and 8.15%. With the same measured single-effect sum, hit rates are 11.45% for the direct head and 8.15% for the removed-input control. SCEPI reaches 11.17%. These controls retain the residual-accuracy contribution of cycle subtraction and the prediction and selection contribution of measured pair inputs.

Table 9. Full-target S-all sensitivity comparison at 25% sequence identity. The columns use the same units and protein cohorts as Table 8. Each predictor is evaluated across five outer partitions and five training repetitions.

Predictor or variant	Phenotype	Residual	Hit rate	Recovery
SCEPI	0.677	0.604	11.15	−0.256
Additive	0.994	1.091	6.65	−0.393
Global nonlinear	0.938	0.650	4.42	−0.432
ThermoMPNN-D	0.758	0.771	6.97	−0.432
ThermoMPNN additive	0.986	1.091	4.95	−0.536
Kermut	0.798	0.847	5.40	−0.458
StabilityArc + Kermut	0.840	0.874	8.23	−0.348
Mean MLP	1.211	0.922	11.20	−0.400
Mean ridge	1.638	1.039	11.25	−0.393
ESM-2 additive	1.173	1.091	2.50	−0.794
ESM-2 context	1.104	0.864	8.50	−0.499
Pair removed	1.151	1.073	4.42	−0.432
Direct residual head	0.699	0.630	11.38	−0.257
Predicted singles	0.855	0.814	8.15	−0.386
Local context	0.664	0.587	11.07	−0.266
Wild-type context	0.655	0.588	10.12	−0.294
Identity features	0.622	0.554	9.82	−0.287
Phenotype loss	0.653	0.584	11.98	−0.235
Residual loss	1.166	0.682	9.77	−0.311
Frozen singles	0.684	0.611	11.32	−0.256
Identity map	0.680	0.609	11.07	−0.252
Structure	0.683	0.609	11.75	−0.240
Permuted distances^{I}	0.685	0.611	11.93	−0.241
Permuted distances^{R}	0.677	0.606	11.75	−0.238
Rewired distances^{I}	0.684	0.609	11.62	−0.243
Rewired distances^{R}	0.688	0.615	10.62	−0.257

G Evaluation on the designated site-pair subset

The separate site-pair evaluation uses the designated unseen-site-pair test subset within each target protein. It intersects this subset with the same held-out homology groups. Site-pair split seeds are 11, 23, 47, 89, and 131. Candidate eligibility can change across these splits. The residual comparison contains 70 proteins, 34 groups, and 339 paired protein-repetition records. Compensation contains 63 proteins, 32 groups, and 269 records. Table 10 reports this subset without combining its values with the full-target results.

Table 10. Repeated S-all evaluation on the designated unseen-site-pair subset within held-out homology groups. Each eligible protein is averaged over its available repetitions.

Predictor	Residual RMSE	Native hit rate (%)	Recovery
SCEPI	0.549	5.75	−0.442
ThermoMPNN-D	0.778	4.88	−0.511
Kermut	0.833	3.29	−0.562
Global nonlinear	0.590	3.56	−0.528
Additive	1.080	4.71	−0.493

With the same measured single-effect sum, top-10 hit rates on this subset are 5.78% for SCEPI, 3.29% for ThermoMPNN-D, 3.07% for Kermut, 3.91% for global nonlinearity, and 3.62% for additivity. At margin 0.5 kcal/mol, the corresponding rates are 5.38%, 3.12%, 2.67%, 3.64%, and 3.34%. That margin cohort contains 62 proteins, 32 groups, and 263 paired protein-repetition records. Homology-group resampling with 10,000 draws and seed 20261010 gives an additive-comparison gain of 2.04 percentage points and a 95% interval of [0.67, 3.31]. Its gain over global nonlinearity is 1.74 points with an interval of [0.49, 3.12]. In S-all, every target double label is withheld from fitting. The site-pair subset defines an additional evaluation population. The target-combination-adaptation experiments use the pair split to test extrapolation from permitted training combinations.

H Target-label budgets and unseen-site-pair prediction

The transferred predictor benefits from partial target single-mutant measurements. Across the same 155 residual-evaluation proteins, its RMSE is 0.793 kcal/mol in Z, 0.788 in S32, 0.772 in S128, and 0.595 in S-all. Table 11 reports these four information conditions. Tables 13–15 report the separate target combination-adaptation experiments.

The family-level comparison varies only the target information condition. Source training and validation remain outside each target homology group. All eligible target double mutants enter evaluation. Each condition retains 155 residual-evaluation proteins and 775 protein-repetition records. Kermut requires target measurements, so the Z comparison lists source-trained predictors.

Table 11. Residual RMSE in kcal/mol at different target single-mutant budgets under 30% homology grouping. Each row covers the same 155 proteins and five training repetitions. Z supplies no target labels; S-all supplies all permitted target single-mutant labels.

Condition	SCEPI	Additive	Global nonlinear	ThermoMPNN-D	Kermut
Z	0.793	1.091	0.861	1.006	—
S32	0.788	1.091	0.857	0.943	0.998
S128	0.772	1.091	0.841	0.908	0.928
S-all	0.595	1.091	0.635	0.805	0.847

The ESM-MSR comparison uses the epoch-10 source-validation checkpoint from family30, fold 0, and repetition seed 11. We compare every method on its matched held-out proteins in that fold. Residual evaluation retains 61 proteins; phenotype evaluation retains 65. Table 12 reports Z and S-all separately. This checkpoint covers one fold and one repetition. Its results remain separate from the complete five-fold, five-repetition comparison.

Table 12. Matched comparison with the available epoch-10 ESM-MSR checkpoint. RMSE uses kcal/mol. Columns report residual RMSE, phenotype RMSE, and phenotype Spearman correlation. Every method shares the same 61 residual proteins and 65 phenotype proteins within each information condition. Kermut requires target labels and is omitted in Z.

	Z			S-all		
Predictor	Residual	Phenotype	Spearman	Residual	Phenotype	Spearman
SCEPI	**0.744**	**1.071**	0.598	**0.550**	**0.621**	**0.767**
ESM-MSR	0.899	1.185	**0.646**	0.891	1.091	0.678
ThermoMPNN-D	1.031	1.237	0.587	0.805	0.772	0.707
Global nonlinear	0.829	1.100	0.534	0.632	0.888	0.735
Additive	1.035	1.136	0.588	1.035	0.885	0.731
Kermut	—	—	—	0.829	0.793	0.731

The C comparison fits the SCEPI adaptation variant to permitted target training combinations. Its source initialization uses an identity phenotype map and a single-effect head held fixed during source pair fitting. Target adaptation retains that head and updates the pair network and observation noise. Target validation selects the stopping point. The target-only variant fits its single-effect head without supervised source weights. Both variants retain the same target single-label permissions. Pair removal sets the fitted pair output to zero at inference. Superscripts S and T denote source initialization and target-only fitting. For mean ridge and mean MLP, S denotes a source prior and T denotes target fitting. Numerical superscripts on MoCHI and MAVE-NN specify interaction order. These adaptation models are evaluated separately from the jointly trained S-all predictor.

Each budget includes proteins with enough permitted training combinations and complete test cycles. Every comparator in a table uses the same protein-repetition intersection. Unseen-site-pair evaluation excludes all test pairs from target combination fitting. Random splitting shares site pairs between training and test combinations. The auxiliary two-pair cohort retains its designated pair split. Coverage reports proteins and matched protein-repetition records. Metrics average repetitions within protein before equal-protein averaging.

Table 13. Unseen-site-pair residual RMSE in kcal/mol after target combination adaptation. Coverage lists proteins and matched protein-repetition records. All methods share the cohort at each budget. Superscripts S and T denote source initialization and target fitting; numerical superscripts give interaction order. C-all uses all permitted training combinations. Test site pairs remain excluded from fitting.

Predictor	C128	C512	C2048	C-all
Coverage	70 / 339	45 / 219	4 / 17	70 / 339
SCEPI adaptation^{S}	0.567	0.570	0.424	0.559
Pair removed^{S}	1.080	1.163	0.967	1.080
SCEPI adaptation^{T}	0.799	0.788	0.495	0.774
Pair removed^{T}	1.080	1.163	0.967	1.080
SCEPI structure^{S}	**0.558**	**0.563**	**0.408**	**0.551**
SCEPI structure^{T}	0.795	0.785	0.467	0.772
ThermoMPNN-D	0.775	0.787	0.618	0.771
ThermoMPNN additive	1.080	1.163	0.967	1.080
Kermut	0.758	0.770	0.577	0.740
MoCHI^{1}	0.857	0.946	0.834	0.900
MoCHI^{2}	0.859	0.935	0.845	0.895
Mean ridge^{T}	0.933	0.994	0.767	0.921
Mean MLP^{T}	0.666	0.635	0.430	0.620
Mean ridge^{S}	1.000	1.067	0.917	0.999
Mean MLP^{S}	0.849	0.882	0.670	0.816

Table 14. Random-split residual RMSE in kcal/mol after target combination adaptation. Every row shares the cohort and averaging rule within each budget. Superscripts follow Table 13. Random splitting evaluates new amino-acid combinations at site pairs represented in training.

Predictor	C128	C512	C2048	C-all
Coverage	125 / 601	90 / 450	11 / 55	125 / 601
SCEPI adaptation^{S}	0.355	0.315	0.279	0.277
Pair removed^{S}	1.065	1.122	1.212	1.065
SCEPI adaptation^{T}	0.401	0.358	0.311	0.311
Pair removed^{T}	1.065	1.122	1.212	1.065
SCEPI structure^{S}	**0.351**	**0.310**	**0.273**	**0.274**
SCEPI structure^{T}	0.399	0.350	0.303	0.304
ThermoMPNN-D	0.424	0.394	0.381	0.385
ThermoMPNN additive	1.065	1.122	1.212	1.065
Kermut	0.447	0.405	0.355	0.346
MoCHI^{1}	0.737	0.802	0.991	0.810
MoCHI^{2}	0.736	0.800	0.997	0.813
Mean ridge^{T}	0.527	0.498	0.568	0.449
Mean MLP^{T}	0.389	0.339	0.283	0.305
Mean ridge^{S}	0.971	1.031	1.124	0.967
Mean MLP^{S}	0.749	0.746	0.726	0.690

Table 15. Residual RMSE in kcal/mol on the auxiliary two-pair cohort. Coverage lists proteins and matched protein-repetition records. Superscripts follow Table 13.

Predictor	C128	C-all
Coverage	27 / 131	29 / 140
SCEPI adaptation^{S}	0.683	0.687
Pair removed^{S}	1.163	1.172
SCEPI adaptation^{T}	1.138	1.149
Pair removed^{T}	1.163	1.172
SCEPI structure^{S}	**0.653**	**0.659**
SCEPI structure^{T}	1.147	1.157
ThermoMPNN-D	0.813	0.823
ThermoMPNN additive	1.163	1.172
Kermut	0.840	0.827
Mean ridge^{S}	1.063	1.074
Mean MLP^{S}	1.002	1.001

I Functional assays and higher-order target adaptation

The functional double-mutant cohort contains 74,598 double mutants from six ProteinGym assays [16]. Measurements retain their original scales. The three KRAS abundance blocks remain separate measurement conditions. The resulting eight conditions contain 7,898 single mutants. Seven conditions supply measured wild-type values and 70,198 complete cycles. HIS7 contributes phenotype-ranking tests and has no observed wild type for cycle evaluation. Table 16 lists the coverage. Each fit uses only its permitted measurement condition.

We evaluate random combination splits and held-out site pairs at C128, C512, C2048, and C-all. Each condition supplies its permitted single mutants and training combinations. Test combinations remain excluded from fitting and validation. The C2048 condition excludes HIS7 because its training pool is smaller than the budget. Phenotype rank correlation is computed on measured test double mutants. Residual evaluation requires at least 30 complete test cycles. To compare assay scales, residual RMSE is divided by the population standard deviation of permitted training point values. Tables report equal-assay means. Repetitions are averaged within condition, the three KRAS conditions are averaged within their assay, and assays receive equal weight.

Table 16. Accepted functional double-mutant data and complete measured cycles. Each row retains its original measurement condition. The three KRAS conditions belong to one abundance assay. HIS7 lacks an observed wild type and contributes no complete cycles.

Assay condition	Singles	Doubles	Complete cycles
D7PM05_CLYGR fluorescence	1,169	10,148	9,517
GFP_AEQVI fluorescence	1,084	12,777	11,954
HIS7_YEAST selection fitness	168	1,475	0
Q6WV12_9MAXI fluorescence	1,141	15,992	15,717
Q8WTC7_9CNID fluorescence	1,201	11,260	10,822
KRAS abundance, block 1	1,188	11,946	11,904
KRAS abundance, block 2	1,122	7,345	7,184
KRAS abundance, block 3	825	3,655	3,100
Total	7,898	74,598	70,198

The functional SCEPI adaptation variant first fits single-effect predictions and a target-specific monotone phenotype map. It then freezes both and fits the pair function. The source-initialized version starts from source models trained with an identity phenotype map and a fixed single-effect head. Related source proteins are excluded using the recorded homology relation. Eligible source models are averaged at the effect level. The target-only version uses no supervised source weights. These variants use predicted single effects within the pair network. They evaluate source initialization and target combination fitting separately from measured-input conditioning in the primary S-all predictor.

The independent no-pair control fits the first stage without adding pair effects. Inference-time pair removal uses the fitted two-stage model and sets its pair output to zero. Learning rates are 10^{-4}, 3×10^{-4}, and 10^{-3}. Target-only models compare projection dimensions 64 and 128. Pair fitting compares observation-loss weights 0.1 and 1. Each stage runs for at most 100 epochs with ten-epoch validation patience. Target validation selects phenotype rank correlation, with identifiable phenotype RMSE breaking ties.

MoCHI and MAVE-NN compare first-order and second-order sequence models [4, 11]. Reference-free analysis (RFA) uses second-order effects with the author's sigmoid phenotype link [15]. The fixed configuration sets $\lambda = 0$. Its optimization budget is 500 L-BFGS iterations, with gradient threshold 10^{-5} and relative objective tolerance 10^{-4} over 25 iterations. Normalization has a separate 500-iteration cap. The validation-selected configuration compares twelve penalties, $\lambda/n \in \{0, 10^{-6}, 10^{-5.5}, \ldots, 10^{-1}\}$. Test outcomes are excluded from fitting and configuration selection.

RFA covers 350 eligible fits, comprising 310 functional-assay fits and 40 FLIP2 fits. Of these, 284 use the fixed configuration and 66 use validation-selected penalties. Thirty additional tasks lack the required sample budget or target training and validation labels and are excluded from fitting.

RFA gives constant GFP predictions in all five C128 repetitions under each split and in two unseen-pair C512 repetitions. Their twelve Spearman correlations are undefined. Rank comparisons use the same finite protein-repetition records for every method. C128 therefore excludes GFP and retains five assays, seven conditions, and 35 records. Unseen-pair C512 retains six assays, eight conditions, and 38 records. The other six-assay rank columns contain 40 records. C2048 retains five assays and 35 records after excluding HIS7. Residual evaluation retains all seven eligible conditions and 35 records at every budget, including the constant-prediction cases.

Tables 17 and 18 separate residual prediction from phenotype ranking. At C128 on unseen pairs, the source-initialized interaction variant has normalized residual RMSE 0.508. Its pair-removed control has 0.551, and the target-only interaction variant has 0.537. First-order MAVE-NN reaches 0.484. At C-all, RFA reaches 0.409 and the source-initialized variant reaches 0.448. On the full six-assay unseen-pair ranking cohort at C128, the source-initialized variant has Spearman correlation 0.730, compared with 0.741 after pair removal. The pair fit improves residual accuracy in this setting while the phenotype-ranking comparison favors the no-pair control.

Table 17. Normalized residual RMSE on functional double mutants. Every column contains five assays, seven measurement conditions, and five repetitions. Errors are divided by each condition's training phenotype standard deviation before equal-assay averaging. Superscripts S and T denote source initialization and target-only fitting; numerical superscripts give interaction order. C-all uses the permitted training combinations.

	Random combinations				Unseen site pairs			
Predictor	C128	C512	C2048	C-all	C128	C512	C2048	C-all
SCEPI adaptation^{S}	0.503	0.484	0.465	0.435	0.508	0.491	0.472	0.448
Pair removed^{S}	0.551	0.529	0.508	0.478	0.551	0.532	0.514	0.488
Independent no-pair^{S}	0.551	0.529	0.508	0.480	0.551	0.532	0.514	0.490
SCEPI adaptation^{T}	0.537	0.515	0.489	0.439	0.537	0.519	0.491	0.459
Pair removed^{T}	0.549	0.527	0.499	0.459	0.549	0.530	0.502	0.469
Independent no-pair^{T}	0.549	0.527	0.499	0.459	0.549	0.530	0.502	0.469
MoCHI^{1}	0.545	0.497	0.453	0.452	0.543	0.499	0.458	0.427
MoCHI^{2}	0.532	0.498	0.453	0.444	0.533	0.497	0.464	0.430
MAVE-NN^{1}	0.476	0.457	0.429	0.407	0.484	0.464	0.431	0.415
MAVE-NN^{2}	0.481	0.460	0.442	0.425	0.484	0.461	0.446	0.437
RFA	0.544	0.524	0.446	0.398	0.548	0.528	0.441	0.409
Mean ridge	0.749	0.687	0.633	0.576	0.792	0.716	0.651	0.610
Mean MLP	0.800	0.715	0.657	0.604	0.793	0.738	0.673	0.619

Table 18. Phenotype Spearman correlation on common finite records for functional double mutants. Every model has five fitting repetitions. Assay counts are given for each budget. Conditions are averaged within assay before equal-assay averaging. Superscripts follow Table 17.

	Random combinations				Unseen site pairs			
Predictor	C128	C512	C2048	C-all	C128	C512	C2048	C-all
Assays	5	6	5	6	5	6	5	6
SCEPI adaptation^{S}	0.739	0.731	0.828	0.771	0.748	0.737	0.827	0.763
Pair removed^{S}	0.754	0.734	0.830	0.767	0.758	0.742	0.829	0.767
Independent no-pair^{S}	0.755	0.734	0.830	0.769	0.755	0.745	0.830	0.767
SCEPI adaptation^{T}	0.716	0.719	0.824	0.766	0.731	0.734	0.824	0.769
Pair removed^{T}	0.720	0.723	0.823	0.767	0.733	0.733	0.824	0.769
Independent no-pair^{T}	0.720	0.723	0.823	0.766	0.732	0.733	0.824	0.769
MoCHI^{1}	0.706	0.711	0.824	0.772	0.707	0.718	0.833	0.760
MoCHI^{2}	0.713	0.727	0.836	0.773	0.721	0.738	0.844	0.766
MAVE-NN^{1}	0.726	0.731	0.830	0.775	0.753	0.741	0.838	0.769
MAVE-NN^{2}	0.726	0.732	0.831	0.769	0.751	0.741	0.840	0.763
RFA	0.705	0.692	0.791	0.747	0.716	0.692	0.798	0.739
Mean ridge	0.602	0.592	0.695	0.662	0.617	0.597	0.691	0.659
Mean MLP	0.557	0.548	0.662	0.649	0.566	0.555	0.661	0.641

FLIP2 evaluates TrpB one-to-many, two-to-many, and by-position splits; amylase one-to-many; IRED two-to-many; Hydro three-to-many; and three Hydro background-transfer splits [17]. We retain the continuous phenotypes and official test assignments. Each split is evaluated separately across five repetitions. Hydro three-to-many averages its three target backgrounds equally. The background-transfer tests provide training measurements from the other backgrounds. Per-background MoCHI, MAVE-NN, and RFA fits require

target-background training labels and are unavailable in those three tests. Pooled representation controls and the adaptation variants use the permitted cross-background training pool.

Tables 19 and 20 report every split. Source-initialized pair fitting improves the TrpB by-position correlation from 0.167 to 0.233. It lowers the one-to-many correlation from 0.252 to −0.214. On Hydro three-to-many, the corresponding values are 0.356 and 0.201. The source-initialized independent no-pair model attains 0.703 on amylase. These results give separate comparisons of source initialization, target fitting, and inference-time pair removal. They establish task-dependent performance for the predicted-single adaptation variant.

Table 19. FLIP2 phenotype Spearman correlation on TrpB, amylase, and IRED. All values use the official test split and average five repetitions. TrpB columns distinguish one-to-many, two-to-many, and by-position protocols. Amylase uses one-to-many; IRED uses two-to-many. Superscripts follow Table 17.

	TrpB			Amylase	IRED
Predictor	1-to-many	2-to-many	By position	1-to-many	2-to-many
SCEPI adaptation^{S}	−0.214	0.155	0.233	0.697	0.200
Pair removed^{S}	0.252	0.172	0.167	0.697	0.221
Independent no-pair^{S}	0.252	0.215	0.172	0.703	0.221
SCEPI adaptation^{T}	0.388	0.206	0.180	0.677	0.222
Pair removed^{T}	0.388	0.413	0.180	0.677	0.219
Independent no-pair^{T}	0.388	0.413	0.180	0.677	0.219
MoCHI^{1}	0.247	0.481	0.256	0.593	0.173
MoCHI^{2}	0.336	0.482	0.218	0.621	0.191
MAVE-NN^{1}	0.245	0.393	0.254	0.574	0.175
MAVE-NN^{2}	0.338	0.460	0.135	0.584	0.175
RFA	0.319	0.180	0.263	0.625	0.181
Mean ridge	0.454	0.466	0.283	0.693	0.199
Mean MLP	0.437	0.485	0.302	0.695	0.173

Table 20. FLIP2 Hydro phenotype Spearman correlation. Three-to-many averages three target backgrounds; each remaining column holds out the indicated background. Every value averages five repetitions. Dashes indicate local fits without target-background training labels. Superscripts follow Table 17.

Predictor	Three-to-many	P01053	P06241	P0A9X9
SCEPI adaptation^{S}	0.201	0.368	0.193	0.025
Pair removed^{S}	0.356	0.420	0.197	0.188
Independent no-pair^{S}	0.356	0.420	0.215	0.188
SCEPI adaptation^{T}	0.059	0.408	0.219	0.036
Pair removed^{T}	0.284	0.408	0.251	0.193
Independent no-pair^{T}	0.284	0.404	0.258	0.209
MoCHI^{1}	0.549	—	—	—
MoCHI^{2}	0.557	—	—	—
MAVE-NN^{1}	0.554	—	—	—
MAVE-NN^{2}	0.565	—	—	—
RFA	0.519	—	—	—
Mean ridge	0.586	0.239	0.025	0.168
Mean MLP	0.614	0.317	0.303	0.025

J Interpretation of the selection endpoint

Complete, point-valued double-mutant cycles define the retrospective evaluation cohort. Their double-mutant outcomes are excluded from model training and candidate scores. Measurement status and distance from assay limits define evaluation strata. The margin is the minimum of $y+1$ and $5-y$ across all four observations. A threshold of 0.5 kcal/mol excludes cycles close to either assay limit. This analysis conditions the evaluation on measured outcomes inside the assay range.

Recovery is $y_{AB} - \max(y_A, y_B)$. A hit requires at least 0.5 kcal/mol recovery, and both singles must individually lose at least 0.5 kcal/mol relative to wild type. Hit rate and mean recovery have different roles: the first measures successful budgeted discoveries, and the second measures continuous candidate quality. A candidate can meet the recovery criterion while remaining less stable than wild type.

General stability ranking is evaluated on a broader continuous pool of 65 proteins in one repetition with seed 11. With the same measured single-effect sum, top-10 mean stability changes differ from additive and global nonlinear ranking by -0.0018 and -0.0054 kcal/mol, respectively.

K Random selection and compensatory enrichment

For protein p, let N_p be the number of eligible candidates and H_p the number satisfying the measured recovery threshold. Uniform sampling of K candidates without replacement gives a hit count $X_p \sim \text{Hypergeometric}(N_p, H_p, K)$. Its expected hit rate is $q_p = H_p/N_p$, and

$$\text{Var}(X_p/K) = \frac{q_p(1-q_p)(N_p-K)}{K(N_p-1)}. \tag{16}$$

Expected mean recovery equals the candidate-pool mean. Both expectations are independent of K. They are calculated exactly from every eligible candidate; no random-ranking simulation is needed.

Let $\overline{h}_p(K)$ be a predictor's hit rate averaged over its five fitted repetitions. For the same P eligible proteins, enrichment is

$$E(K) = \frac{|P|^{-1}\sum_{p\in P}\overline{h}_p(K)}{|P|^{-1}\sum_{p\in P} q_p}. \tag{17}$$

This ratio uses the same equal-protein aggregation as the primary results. It remains defined when individual proteins have no observed hits. The standard cohort contains 79,564 candidates and 1,718 hits across 120 proteins. Its pooled candidate hit fraction is 2.16%, and its equal-protein mean random hit rate is 2.63%. The latter is the denominator for every reported enrichment.

The random comparison uses the same fixed model predictions, candidate eligibility, recovery threshold, budgets, and measurement margins as the other selection comparisons. Each predictor's five repetitions are averaged within protein before aggregation. Uncertainty uses 10,000 paired homology-group bootstrap draws with seed 20261019. Each draw resamples model and random results together and recalculates both means and their ratio. These intervals quantify sampling uncertainty in the observed cohort. The exact random expectation has no Monte Carlo ranking error.

Table 21. SCEPI native selection relative to exact uniform random selection under 30% homology grouping. The same 120 proteins and 44 groups enter every budget. Random hit rate is 2.63%. Hit-rate gains are percentage points; recovery gains are kcal/mol. Brackets give 95% homology-group bootstrap intervals for enrichment.

Budget	SCEPI hit rate (%)	Hit-rate gain	Enrichment (×)	Recovery gain
1	14.50	11.87	5.52 [2.58, 8.24]	0.784
3	12.89	10.26	4.90 [2.59, 7.11]	0.728
10	11.22	8.59	4.27 [2.74, 5.89]	0.643
30	8.56	5.93	3.26 [2.43, 4.40]	0.529

At $K = 10$, SCEPI's hit-rate gain over random selection has a 95% interval of [4.01, 12.07] percentage points. Its recovery gain has an interval of [0.499, 0.720] kcal/mol. The stricter homology grouping gives 4.24-fold native enrichment. A margin of 0.5 kcal/mol retains 115 proteins and 75,017 candidates. Native enrichment is 4.15 and 4.19 under the 30% and 25% groupings. With the same measured single-effect sum, enrichment is 4.17 and 4.20 in these cohorts (Table 22).

Table 22. Top-10 random-selection sensitivity. Random rates and model rates are equal-protein means. The last two columns report SCEPI enrichment with native predictions and the fixed measured single-effect sum. Both use the same random expectation. The 30% grouping has 44/44/43 groups at the three margins; the 25% grouping has 39/39/38.

Grouping	Margin	Proteins	Random hit (%)	Native (×)	Fixed sum (×)
30%	0.00	120	2.63	4.27	4.29
30%	0.25	120	2.68	4.17	4.19
30%	0.50	115	2.68	4.15	4.17
25%	0.00	120	2.63	4.24	4.25
25%	0.25	120	2.68	4.17	4.17
25%	0.50	115	2.68	4.19	4.20